

Planning Wizard - Calculation Formulas

This document contains all the formulas used in the Planning Wizard for technical sizing, economic analysis, and emissions calculations.

Technical Recommended Rating FORMULAS

1.1 Basic Power Calculations

Average Power (kW)

Average Power = Daily Consumption (kWh) / 24 hours

Peak Power (kW)

Peak Power = Average Power × Peak-to-Average Ratio

Where Peak-to-Average Ratio:

- Residential: 2.5
- Commercial: 1.8
- Industrial: 1.3

Annual Consumption (kWh)

Annual Consumption = Daily Consumption × 365 days

1.2 Solar Capacity Sizing

Solar Capacity (kW)

Solar Capacity = Daily Consumption / (Peak Sun Hours × System Efficiency)

Where:

- Peak Sun Hours = 5 hours/day
- System Efficiency = 0.773 (77.3%)

Daily Solar Generation (kWh)

Daily Solar Generation = Solar Capacity × Peak Sun Hours × System Efficiency

1.3 Battery Capacity Sizing

Battery Capacity (kWh)

Battery Capacity = Daily Consumption × Battery-to-Daily Ratio

Where Battery-to-Daily Ratio:

- Residential: 0.83 (83%)
- Commercial: 0.65 (65%)
- Industrial: 0.55 (55%)

Battery Capacity (Ah)

$$\text{Battery Capacity (Ah)} = (\text{Battery Capacity (kWh)} \times 1000) / \text{Battery Voltage (V)}$$

Where Battery Voltage:

- Residential: 12V
- Commercial: 48V
- Industrial: 480V

Backup Hours

$$\text{Backup Hours} = \text{Battery Capacity (kWh)} / \text{Average Power (kW)}$$

1.4 Inverter & DC-DC Converter Sizing

Inverter Capacity (kW/kVA)

$$\text{Inverter Capacity} = \text{Solar Capacity} \times 1.25$$

DC-DC Converter Rating (kW)

$$\text{DC-DC Converter Rating} = \text{Solar Capacity} \times 1.25$$

1.5 Grid & Diesel Sizing

Grid Connection (kW)

$$\text{Grid Connection} = \text{Peak Power (kW)}$$

Diesel Capacity (kW)

$$\text{Diesel Capacity} = \text{Peak Power} \times 1.1$$

1.6 Hydrogen System (Optional)

Electrolyzer Capacity (kW)

$$\text{Electrolyzer Capacity} = \text{Solar Capacity} \times 0.2 \text{ (20\% of solar)}$$

Fuel Cell Capacity (kW)

$$\text{Fuel Cell Capacity} = \text{Peak Power} \times 0.5 \text{ (50\% of peak)}$$

H2 Tank Capacity (kg)

$$\text{H2 Tank Capacity} = \text{Battery Capacity (kWh)} / 33.3$$

Where: 1 kg H2 $\approx$ 33.3 kWh

2. ECONOMIC ANALYSIS FORMULAS

2.1 Cost Assumptions

Unit Costs:

- Solar Cost: ₹40,000 per kW
- Battery Cost: ₹8,000 per kWh
- Inverter Cost: ₹8,000 per kW
- DC-DC Converter Cost: ₹1,500 per kW
- Installation: 10% of equipment cost
- O&M: 3% of (Solar + Battery) cost per year

Hydrogen System Costs (if applicable):

- Electrolyzer: ₹50,000 per kW
- Fuel Cell: ₹60,000 per kW
- H2 Tank: ₹5,000 per kg

2.2 Dual Mode System Costs

Equipment Costs:

Solar Cost = Solar Capacity × ₹40,000
Battery Cost = Battery Capacity × ₹8,000
Inverter Cost = Inverter Capacity × ₹8,000
DC-DC Converter Cost = DC-DC Converter Rating × ₹1,500

Installation Cost:

Installation Cost = (Solar + Battery + Inverter + DC-DC Converter) × 0.10

Total CAPEX:

Total CAPEX = Equipment Cost + Installation Cost

Annual O&M Cost:

Annual O&M = (Solar Cost + Battery Cost) × 0.03
Monthly O&M = Annual O&M / 12

2.3 On-Grid System Costs

Equipment Costs:

Solar Cost = Solar Capacity × ₹40,000
Inverter Cost = Inverter Capacity × ₹8,000
Battery Cost = 0 (No battery)
DC-DC Converter Cost = 0 (No DC-DC converter)

Installation Cost:

Installation Cost = (Solar + Inverter) × 0.10

Total CAPEX:

Total CAPEX = Equipment Cost + Installation Cost

Annual O&M Cost:

Annual O&M = Solar Cost $\times$ 0.03

Monthly O&M = Annual O&M / 12

2.4 Annual Generation & Revenue

System Efficiencies:

- Dual Mode System Efficiency: 0.80 (80%) - lower due to battery round-trip losses
- On-Grid System Efficiency: 0.85 (85%) - higher, direct grid connection

Annual Solar Generation:

Dual Mode: Annual Generation = Solar Capacity $\times$ 5 $\times$ 365 $\times$ 0.80

On-Grid: Annual Generation = Solar Capacity $\times$ 5 $\times$ 365 $\times$ 0.85

Grid Tariffs:

- Grid Export Tariff: ₹8.0 per kWh (selling excess solar)
- Grid Purchase Tariff: ₹8.5 per kWh (buying from grid)

Annual Revenue:

Dual Mode Revenue = Annual Generation (Dual Mode) $\times$ ₹8.0

On-Grid Revenue = Annual Generation (On-Grid) $\times$ ₹8.0

Grid Energy Needed:

Dual Mode Grid Energy = $\max(0, \text{Annual Consumption} - \text{Annual Generation (Dual Mode)})$

On-Grid Grid Energy = $\max(0, \text{Annual Consumption} - \text{Annual Generation (On-Grid)})$

Annual Grid Cost:

Dual Mode Grid Cost = Dual Mode Grid Energy $\times$ ₹8.5

On-Grid Grid Cost = On-Grid Grid Energy $\times$ ₹8.5

2.5 Annual Savings & Payback Period

Annual Savings:

Dual Mode Savings = Annual Revenue - Annual Grid Cost - Annual O&M

On-Grid Savings = Annual Revenue - Annual Grid Cost - Annual O&M

Monthly Savings:

Monthly Savings = Annual Savings / 12

Payback Period (years):

$\text{Payback Period} = \text{Total CAPEX} / \text{Annual Savings}$

(If Annual Savings ≤ 0 , Payback Period = 0)

3. CARBON EMISSIONS CALCULATION FORMULAS

3.1 Emission Factors

Emission Factors:

- Grid Emission Factor: 0.82 kg CO₂ per kWh (Indian grid average)
- Solar Operational Factor: 0.05 kg CO₂ per kWh (operational emissions)
- Solar Manufacturing Factor: 0.04 kg CO₂ per kWh capacity (manufacturing)
- Battery Manufacturing Factor: 150 kg CO₂ per kWh capacity (manufacturing)
- System Lifetime: 25 years

3.2 Lifetime Emissions Calculation

Lifetime Consumption:

$\text{Lifetime Consumption} = \text{Annual Consumption} \times 25 \text{ years}$

Dual Mode System - Lifetime Emissions (25 years):

1. Solar Manufacturing Emissions (one-time):

$\text{Solar Manufacturing} = \text{Solar Capacity (kW)} \times 0.04 \times 1000$

2. Solar Operational Emissions:

$\text{Solar Operational} = \text{Annual Solar Generation (Dual Mode)} \times 0.05 \times 25 \text{ years}$

3. Grid Emissions:

$\text{Grid Emissions} = \text{Dual Mode Grid Energy Needed} \times 0.82 \times 25 \text{ years}$

4. Battery Manufacturing Emissions (one-time):

$\text{Battery Manufacturing} = \text{Battery Capacity (kWh)} \times 150$

5. Total Dual Mode Lifetime Emissions:

$\text{Total Emissions (kg)} = \text{Solar Manufacturing} + \text{Solar Operational} + \text{Grid Emissions} + \text{Battery Manufacturing}$

$\text{Total Emissions (tonnes)} = \text{Total Emissions (kg)} / 1000$

On-Grid System - Lifetime Emissions (25 years):

1. Solar Manufacturing Emissions (one-time):

$\text{Solar Manufacturing} = \text{Solar Capacity (kW)} \times 0.04 \times 1000$

2. Solar Operational Emissions:

Solar Operational = Annual Solar Generation (On-Grid) × 0.05 × 25 years

3. Grid Emissions:

Grid Emissions = On-Grid Grid Energy Needed × 0.82 × 25 years

4. Total On-Grid Lifetime Emissions:

Total Emissions (kg) = Solar Manufacturing + Solar Operational + Grid Emissions

Total Emissions (tonnes) = Total Emissions (kg) / 1000

3.3 Annual Emissions (for display)

Annual Emissions:

Dual Mode Annual Emissions = Dual Mode Lifetime Emissions / 25

On-Grid Annual Emissions = On-Grid Lifetime Emissions / 25

3.4 Carbon Offset Calculation

Pure Grid Baseline:

Pure Grid Annual Emissions = Annual Consumption × 0.82

Pure Grid Lifetime Emissions = Lifetime Consumption × 0.82

Dual Mode Operational Emissions (Annual):

Dual Mode Operational Annual = (Solar Operational + Grid Emissions) / 25 years

Note: Excludes manufacturing emissions for carbon offset calculation

Annual CO₂ Reduction:

Annual CO₂ Reduction = Pure Grid Annual Emissions - Dual Mode Operational Annual Emissions

Carbon Offset Percentage:

Carbon Offset % = (Annual CO₂ Reduction / Pure Grid Annual Emissions) × 100

Lifetime CO₂ Reduction:

Lifetime CO₂ Reduction (tonnes) = (Pure Grid Lifetime Emissions - Dual Mode Lifetime Emissions) / 1000

4. EXAMPLE CALCULATION

Input:

- Daily Consumption: 9.90 kWh/day

- Use Case: Residential

Step 1: Technical Sizing

Average Power = $9.90 / 24 = 0.4125$ kW
Peak Power = $0.4125 \times 2.5 = 1.03125$ kW
Solar Capacity = $9.90 / (5 \times 0.773) = 2.56$ kW
Battery Capacity = $9.90 \times 0.83 = 8.217$ kWh ≈ 8.21 kWh
Inverter Capacity = $2.56 \times 1.25 = 3.2$ kW ≈ 3.19 kW
DC-DC Converter = $2.56 \times 1.25 = 3.2$ kW ≈ 3.19 kW

Step 2: Economic Analysis

Solar Cost = $2.56 \times ₹40,000 = ₹102,400$
Battery Cost = $8.21 \times ₹8,000 = ₹65,680$
Inverter Cost = $3.19 \times ₹8,000 = ₹25,520$
DC-DC Converter Cost = $3.19 \times ₹1,500 = ₹4,785$
Installation = $(102,400 + 65,680 + 25,520 + 4,785) \times 0.10 = ₹19,838.50$
Total CAPEX = ₹217,223.50

Step 3: Annual Generation

Annual Consumption = $9.90 \times 365 = 3,613.5$ kWh
Dual Mode Annual Generation = $2.56 \times 5 \times 365 \times 0.80 = 3,737.6$ kWh
On-Grid Annual Generation = $2.56 \times 5 \times 365 \times 0.85 = 3,972.8$ kWh

Step 4: Carbon Emissions

Pure Grid Annual Emissions = $3,613.5 \times 0.82 = 2,963.07$ kg CO₂
Dual Mode Operational Annual = (Solar Operational + Grid Emissions) / 25
Carbon Offset % = $(\text{Reduction} / 2,963.07) \times 100$

5. NOTES

1. **System Efficiency (77.3%):** This accounts for inverter losses, DC-DC converter losses, and battery round-trip efficiency in the solar sizing calculation.
2. **Battery Sizing:** The battery-to-daily ratio is conservative to account for depth of discharge (DoD) limitations and efficiency losses.
3. **Carbon Offset:** Uses operational emissions only (excludes manufacturing) to match the old website's calculation methodology.
4. **Payback Period:** Calculated as simple payback (CAPEX / Annual Savings), not discounted payback.
5. **Grid Tariffs:** These are default values and can be adjusted based on location and utility rates.

Last Updated: Based on `backend/routes/planning.js` implementation